ORIGINAL ARTICLE

A temporal–spatial deep learning framework leveraging dynamic 3D attention maps for violence detection

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Abstract

In intelligent systems for real-time security and safety monitoring, the proliferation of surveillance cameras has fueled a growing interest in using deep learning-based artificial intelligence (AI) models for violence detection. Most current approaches consider violence detection as a video classification task, overlooking the fact that violent activities occur within relatively small spatiotemporal regions. Moreover, these activities depend on relationships among multiple such regions, making a single region analysis inadequate, especially for larger-scale violence. This paper proposes a novel temporal–spatial attention framework inspired by human visual perception, which dynamically focuses on multiple informative regions across space and time. By learning where, when, and for how long to attend within a video, using dynamic three-dimensional attention prediction networks, the model captures complex patterns of violent behavior more effectively. Experiments on four public benchmark datasets and a real-world dataset created for this study demonstrate that the proposed approach outperforms existing methods in accuracy and interpretability.

Keywords Video surveillance · Violence detection · Computer vision · 3D spatiotemporal attention maps · Residual convolutional neural network

1 Introduction

Video violence detection is a major and simultaneously arduous computer vision problem whose goal is to correctly and efficiently identify the presence of violent events in video material. On the one hand, recent advancements in video technology have opened up opportunities for practical applications of violence detection. These applications include intelligent surveillance systems in natural scenarios and online content review. However, on the other hand, violence detection remains a formidable challenge. This is due to inherent complexities, such as the diverse nature of violent actions and the limited availability of annotated violence data [1].

Traditional surveillance systems often depend on human operators, which can lead to issues such as personnel shortages and missed threats. Research has shown that human operators can fail to detect objects or activities in closed-circuit television (CCTV) footage after just 20 min of monitoring [2, 3]. It points out the need for innovation in automated systems, and over the years, several types of research [4–10] have contributed to violence detection. Early violence detection tasks [4, 5] were primarily focused on short video classification. In such tasks, it was assumed that non-violent and violent videos were well-trimmed, with violent videos containing only violent events. However, such a setting limited the algorithm's application area to short video classification, not generalizing performance in long untrimmed videos for violence detection. Thus, these state-of-the-art methods have their shortcomings regarding practical applications. Then, a few studies have attempted to localize

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violence in untrimmed videos, such as the violent scene detection task [6] and the fighting detector [10]. However, these approaches relied on frame-level annotations, which ignored fine-grained visual cues and spatiotemporal properties of videos. This often led to misclassifications of violence, especially in complex and crowded scenes. Hence, there emerged a need for visual attention mechanisms similar to those used by humans for fine-grained video analysis in violence classification.

The principle of human visual attention is that, with respect to human perception, there is one key characteristic: It is not necessary for people to view and process a whole visual scene all at once. On the contrary, humans selectively direct their attention to portions of the visual space where details may be acquired when needed. They then combine particulars from multiple fixations in due course to form a meaningful scene representation, which is subsequently used for decision-making and perception [11]. In image processing and computer vision applications, attentional models have played a similar role, especially in those tasks for which interpretation or explanation depends on just a small fraction of a video or image. Examples include human action recognition [12, 13], image recognition [14], visual question answering [15, 16], and machine translation [17].

Even though these models offer a degree of interpretability by visualizing the regions they attend to for specific tasks or decisions, they often fall short when it comes to understanding the three-dimensional (3D) spatial relationships crucial for recognizing complex actions and scenes. One important challenge in violence detection, especially in surveillance camera footage, is that the violent actions to be detected usually fall into a very small spatiotemporal region of the video. To address this gap, this paper proposes a deep learning model that integrates multi-attention mechanisms to dynamically attend to critical spatial and temporal segments across video frames.

1.1 Motivation

Figure 1 illustrates why a multi-attention temporal–spatial approach is essential for fine-grained violence detection. In Fig. 1a, a *Fighting* incident occurs at the corner of the frame. Due to its peripheral location and limited spatial spread, it is misclassified as a *Natural* scene. Similarly, Fig. 1b shows a crowded scenario where violent behavior is visible only at the margins of the frame, yet it is incorrectly identified as a *Peaceful Gathering*. These examples highlight the importance of localized attention to fine-grain, small-scale cues that are easily overlooked by global or frame-level models.

These examples emphasize several key challenges in accurately recognizing violence within videos. First, violent actions often occur in spatially marginal regions of the frame or are temporally brief, making them easy to overlook. In crowded or chaotic scenes, such actions may be partially occluded or visually cluttered, further complicating detection. In addition, relying on a single region or frame-level analysis fails to capture the broader

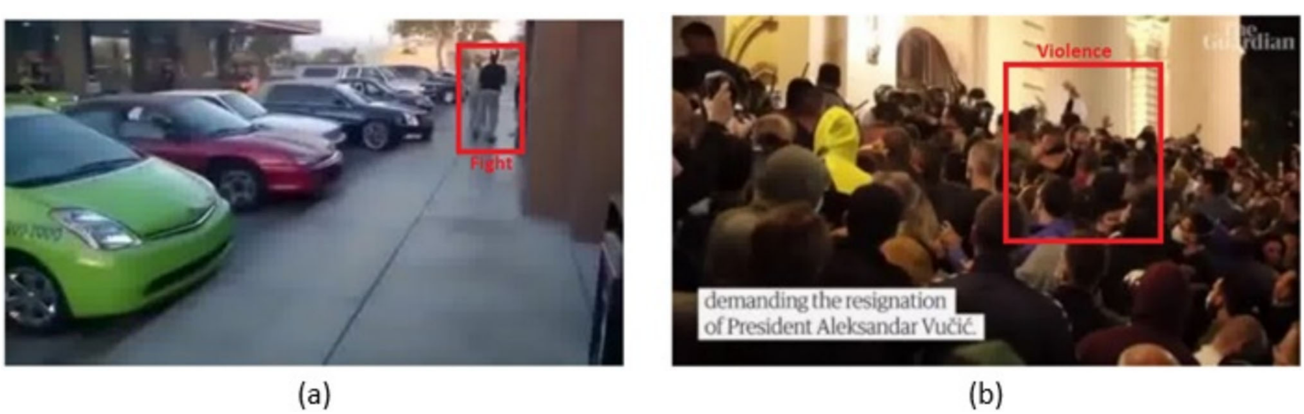

Fig. 1 Sample frames to show the need for a multi-attention temporal–spatial model for fine-grained recognition. **a** The behavior of *Fighting* is at the corner of the frame, which misclassifies the frame as *Natural*. **b** The violent nature of the crowd is visible only at the periphery of the scene and classifies the scene as a *Peaceful Gathering* instead of a *Violent Gathering*

context needed for understanding complex interactions and the necessary cues that signify violence. These limitations trace to the need for a more attuned strategy capable of identifying and processing fine-grained spatiotemporal information.

To overcome these challenges, this paper introduces a dynamic 3D attention-based framework specifically designed for fine-grained violence detection. The proposed model learns to attend to multiple spatiotemporal volumes—3D regions that span spatial and temporal dimensions—within a video sequence. By doing so, it is capable of identifying and isolating critical areas where violent behavior occurs. Moreover, it effectively integrates features from globally extracted context and localized attention regions to enhance the recognition process. Importantly, the model is inspired by human visual attention, dynamically determining where to focus, when to attend, and how long to observe specific segments of the video. This adaptive mechanism significantly improves the model's ability to recognize violence in diverse and complex scenarios. The main contributions of this work are:

- The challenges in fine-grained video recognition are tackled by introducing a new dynamic 3D temporal–spatial framework by adapting the property of human visual attention. This framework supports not only precise identification of distinctive 3D region volumes in a mutually consistent way but also improves learning of spatiotemporal region-based representations.
- The proposed model is designed to provide the necessary information, allowing us to consider where, when, and for how long the model focuses within the video to make determinations regarding violent behavior.
- Experiments were performed on four benchmark datasets and one real-world dataset curated by ourselves. The results consistently reveal better performance of our model compared with related existing approaches.

The remainder of the article is structured as follows: The literature review is presented in Sect. 2; then, the details of the proposed model are outlined in Sect. 3; experiments and outcomes are discussed in Sect. 4; and the article concludes in Sect. 5.

2 Related work

In the realm of video recognition, visualizing the exact parts of each frame and identifying specific frames that the model focuses on provides crucial insights into the model's behavior and decision-making processes [18]. The authors in [19] discussed an attention-driven long short-term memory (LSTM) that emphasizes vital spatial locations for action recognition. In [20], the authors introduced an attention mechanism combining bottom-up and top-down attention, employing bilinear pooling techniques based on low-rank approximations. Nevertheless, these approaches primarily concentrate on identifying the critical spatial locations within individual images, they fail to consider the temporal relations that prevail among successive video frames.

The authors of [21] and [22] integrated visual attention into the motion stream as a temporal attention scheme. However, the existing motion stream was mainly based on optical flow patterns from two adjacent frames and was not able to identify the long-range temporal relationships within clips. Besides, extra optical flow frames were given to the motion stream, which could bring very high overhead to optical flow computation, extraction, and storage, especially on large-scale datasets. Li et al. [23] and Torabi et al. [24] put forward an LSTM model based on attention to emphasize significant video frames. However, here temporal attention was considered without considering any spatial information.

Meanwhile, an end-to-end attention-based temporal and spatial model was proposed by Song et al. [13] for human action recognition; however, it necessitates additional skeleton data. Skeleton image sequences were also utilized by Yang et al. [25] for action recognition that employed a spatiotemporal attention network composed of ResNet 50 residual blocks. The utilization of skeleton data for recognition in these methods often hinders the attainment of fine-grained recognition, primarily due to the limitations associated with inaccurate and incomplete pose estimation.

Recently, attention-based models were used for fine-grained recognition in images [14, 26, 27] and videos [28]. The approach in [14] utilized a recurrent attention CNN (RA-CNN) with an attention proposal network for image recognition in a fine-grained manner, while in [26] and [27], an attention transformer was employed for the same task. Wang et al. [28] employed MobileNet in conjunction with a gated recurrent unit (GRU) to identify patches within video frames that help focus on relevant areas of activity. It is worth noting that the latter methods are specifically designed for images, while the former randomly selects frames without considering the natural temporal properties existing in video data.

Furthermore, deep learning approaches have become extremely popular for violence detection owing to their outstanding performance. Sudhakaran et al. [8] and Hanson et al. [9] leveraged convolutional LSTMs to capture spatiotemporal features. Peixoto et al. [29] utilized dual-stream DNNs to extract high-level violence patterns, while Singh et al. [30] applied deep networks for drone-based surveillance. More recently, Graph Convolution Network (GCN)-based approaches [31] were proposed for modeling interactions via 3D skeleton point clouds.

Even with such advancements, the majority of existing techniques do not comprehend complex 3D spatial relationships or identify violent actions that occupy small, localized volumes in extended, untrimmed videos. Many models utilize spatial or temporal attention in isolation, are not scale invariant, or require additional modalities Like skeletons or optical flow. To overcome all these gaps, our work introduces an attention framework that encodes spatial and temporal relationships by learning dynamic 3D attention maps, eliminating auxiliary data. The forthcoming sections elaborate on the details of the proposed framework and provide a comprehensive analysis of our experiments.

3 Methodology

In this section, we present the proposed model for violent behavior recognition in crowd videos. The architecture integrates spatial–temporal attention, multi-path feature extraction, and feature volume aggregation for effective classification. The overall framework consists of the following components: global feature extraction, attention prediction networks (APNs), local patch extraction, feature fusion, and classification. The architecture of the proposed network is given in Fig. 2.

Let the input to the network be a batch of video clips $\mathbf{X} \in \mathbb{R}^{B \times T \times C \times H \times W}$, where B is the batch size, T is the number of temporal segments, C is the number of channels, and H, W are the frame Height and width, respectively. Each video is normalized using standard mean and standard deviation values. To obtain global-level contextual features, the input video is processed through a 3D CNN backbone pretrained on the Kinetics-400 dataset. The backbone captures long-range spatial–temporal patterns and outputs a global feature vector $\mathbf{f}_g$ for each clip:

$$\mathbf{f}_g = f_G(\mathbf{X}) \tag{1}$$

where f_G is the backbone convolution function to extract the global features. In particular, f_G is a global residual CNN having spatial and temporal modeling capability. The f_G is designed to rapidly analyze each frame, giving additional information for deciding the region for the final classification of violent behavior.

3.1 Dynamic 3D attention prediction network (APN)

To identify implicit and local human behaviors in videos, our framework employs several APNs that dynamically look for informative spatiotemporal areas. Each APN identifies a small cuboid of the video where significant activity is occurring, extracts this region, and processes it separately. These localized regions are merged with

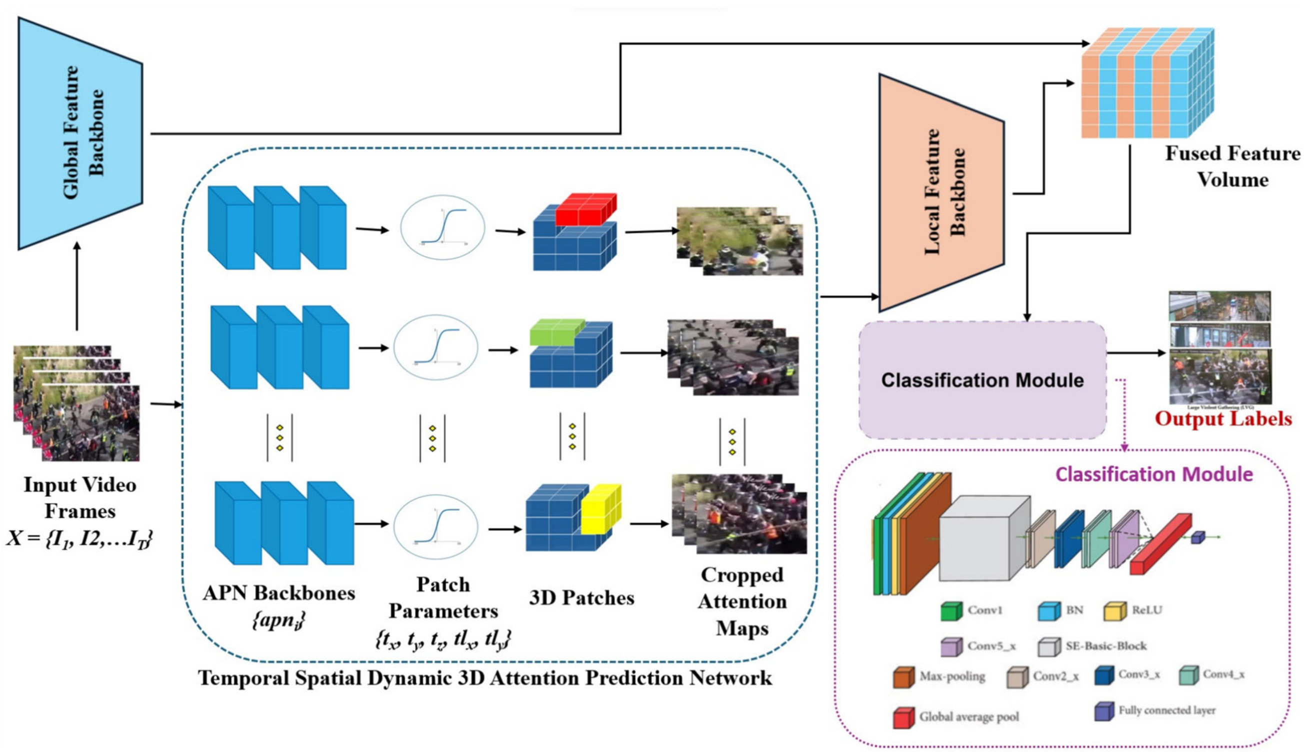

Fig. 2 Architecture of the proposed network. It accepts input video frames and utilizes a global feature backbone to obtain contextual long-range temporal and spatial features. Fine-grained recognition is achieved by dynamic *apn*s using the 3D patches generated from the APN backbones. Finally, the classifier module takes the fused feature volume to predict the output labels

global video features to create a single representation of the scene. This kind of architecture allows the model to attend to multiple regions of interest simultaneously, boosting its ability to see fine-grained activities.

We introduce a 3D dynamic temporal–spatial APN to achieve fine-grained recognition, which takes the video tensor X as input and generates 3D volumes of attention maps. We employ n dynamic APN backbones because a single attention map may not adequately capture the rich information present in various parts of the video. At the outset, each backbone apn_i takes X to predict the spatiotemporal locations of regions of interest. Each apn_i is implemented as a 3D CNN followed by two sequential sigmoid-activated linear layers that regress five parameters: $(t_x, t_y, t_z, t_{lx}, t_{ly})$ corresponding to the patch center coordinates and half-widths in spatial and temporal dimensions. As such, the attended region is approximated as a patch cuboid with given depth d, where (tx_i, ty_i, tz_i) represent the center coordinates of the i^{th} *apn* cuboid in terms of the axes x, y, and z, respectively, while (tlx_i, tly_i) denotes the cuboid's length and breadth. The predicted coordinates are scaled with respect to the input dimensions and clamped to ensure valid cropping boundaries.

3.1.1 Patch extraction

Given the spatial–temporal attention coordinates $t_x, t_y, t_z \in [0, 1]$ predicted by the apn_i, and the normalized patch half-sizes tl_x and tl_y, we extract 3D patches from the input video tensor. The normalized coordinates are first scaled to the actual frame dimensions as follows:

$$
\begin{aligned}
& t'_x = \lfloor W \cdot t_x \rfloor, \quad t'_y = \lfloor H \cdot t_y \rfloor, \quad t'_z = \lfloor d + (T - 2d) \cdot t_z \rfloor \\
& tl'_x = \left\lfloor \frac{W}{2} \cdot tl_x \right\rfloor + 1, \quad tl'_y = \left\lfloor \frac{H}{2} \cdot tl_y \right\rfloor + 1
\end{aligned} \tag{2}
$$

Here, W and H are the width and height of the video frames, T is the total number of frames, and d is the temporal depth for each patch. The spatial–temporal boundaries of each patch are computed as:

$$
\begin{aligned}
& x_{\min} = \max(0, t'_x - tl'_x), \quad x_{\max} = \min(W, t'_x + tl'_x) \\
& y_{\min} = \max(0, t'_y - tl'_y), \quad y_{\max} = \min(H, t'_y + tl'_y) \\
& z_{\min} = t'_z - d, \quad z_{\max} = t'_z + d
\end{aligned} \tag{3}
$$

A 3D patch $\mathbf{P} \in \mathbb{R}^{2d \times C \times h \times w}$ is extracted from the video tensor, where C is the number of channels. Each extracted patch is resized to a fixed resolution of 224×224 using bilinear interpolation to ensure uniformity across samples:

$$
\mathbf{P} = \text{Interpolate}(I[z_{\min} : z_{\max}, :, y_{\min} : y_{\max}, x_{\min} : x_{\max}], \ \text{size} = (224, 224)) \tag{4}
$$

All patches are stacked along the batch dimension to form the final patch tensor used for further processing in the local feature pathway. Algorithm 1 provides a concise explanation of the APN, and Fig. 3 demonstrates the schematic representation of the generation of the attention maps.

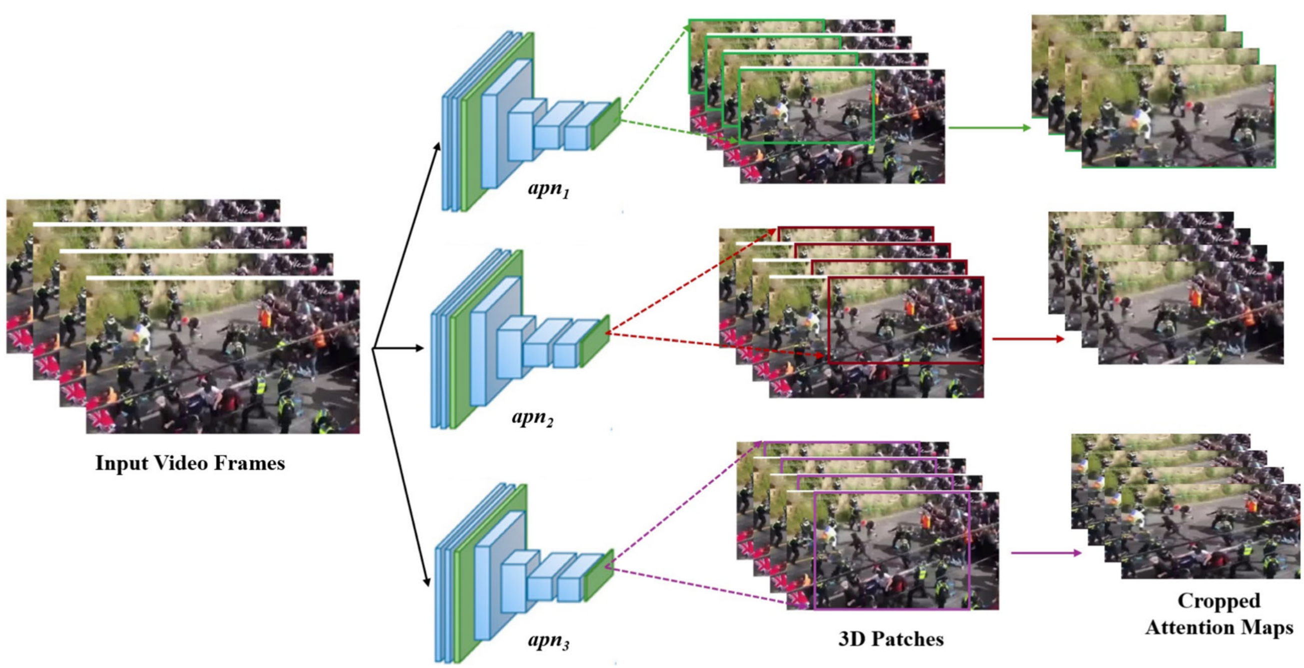

Fig. 3 Schematic representation of the generation of attention maps for $n = 3$ and $d = 4$

Algorithm 1 Dynamic 3D Attention Prediction and Patch Extraction

Require: Video batch $\mathbf{X} \in \mathbb{R}^{B\times T\times C\times H\times W}$
Ensure: Extracted 3D patches $\{\mathbf{P_i}\}$
procedure APN($\mathbf{X}$)
  for each *apn* backbone apn_i, $i = 1, 2, \ldots, n$ **do**
    Predict patch parameters, $(t_x, t_y, t_z, tl_x, tl_y) \leftarrow apn_i(\mathbf{X})$
    Scale normalized center and half width coordinates of the patch to (t'_x, t'_y, t'_z) and $(tl'_x, tl'_y))$ using equation (2)
    Compute temporal spatial patch boundaries, $\mathbf{P_b} \leftarrow [x_{min}, x_{max}, y_{min}, y_{max}, z_{min}, z_{max}]$, using equation (3)
    Crop attention maps, $\mathbf{P} \leftarrow \mathbf{X}(\mathbf{P_b})$.
    Resize $\mathbf{P}$ using equation (4)
  end for
  Stack all patches $\{\mathbf{P_i}\}$ along batch dimension.
end procedure

3.1.2 Feature volume construction

Each cropped patch is passed through a dedicated local feature backbone, ${f_L}^i$, to extract local features:

$$\mathbf{f}_l{}^i = {f_L}^i(\mathbf{P}_i), \quad i \in \{1, 2, \ldots, n\} \tag{5}$$

where n is the number of *apn*s. The global and local features are combined to form a volumetric feature representation. Specifically, we define a 3D tensor $\mathbf{V} \in \mathbb{R}^{B\times N\times N\times N}$, where each voxel is a fusion of the global and local features:

$$\mathbf{V}_{b,i,j,k} = \frac{1}{n+1}\left(\mathbf{f}_1^{(b,i)} + \mathbf{f}_2^{(b,j)} + \mathbf{f}_g^{(b,k)}\right) \tag{6}$$

This ensures permutation invariance and multi-path fusion by populating symmetric permutations of (i, j, k) with the same fused value.

3.2 Training and behavior classification

The volumetric tensor $\mathbf{V}$ is passed through a 2D convolutional layer followed by a pretrained ResNet-18 for final classification. A fully connected head maps the final feature vector to class logits. The categories of predicted classes can be customized based on user requirements, such as no-fight/fight, non-violence/violence, and natural/peaceful gathering/violent gathering. The model undergoes end-to-end training on the training set by performing categorical cross-entropy loss minimization using the Adam optimizer with a fixed learning rate. All subnetworks, including the APNs, feature extractors, and classifier, are trained end-to-end. Batch size, number of segments T, and patch temporal depth d are configurable hyperparameters.

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4 Experimental analysis and results

4.1 Datasets

4.1.1 Multi-scale violence (MSV) dataset

The proposed approach for fine-grained video recognition was tested on datasets of violence or fight-related events since their detection and response at any part of a video are of paramount importance to enhance security in public surveillance systems. With our particular focus on the aspect of crowd behavior understanding with respect to dynamics and levels of violence, we noticed that existing benchmark datasets only differentiate between events that are non-violent and violent. To this end, we have worked on designing a novel and unique dataset, namely the multi-scale violence (MSV) dataset. This dataset was designed to include most crowd categories, given the number of people and the level of violence that exists.

We have organized crowd behaviors into four unique categories to capture specific crowd dynamics and levels of violence. They are: *Fighting (F), Natural (N), Large Violent Gathering (LVG),* and *Large Peaceful Gathering (LPG)*. F stands for small groups that get into physical violence; LPG stands for large, peaceful gatherings such as the audience of a sports stadium or rally for a cause. In contrast, LVG is large crowds acting violently, property destruction, or police battles. All activities not falling into the above category are labeled as N. The classes and the sample frames are given in Fig. 4.

Data collection: The majority of the videos were taken from YouTube. The videos were selected to ensure that they contain at the minimum one occurrence of any of the pertinent classes. The keywords used to crawl the videos are violence, demonstration, and clash. Videos of important historical events, such as *The Capitol riot, Hong Kong protests, and George Floyd protests*, were also searched and included in the dataset. We also included 216 fight-labeled videos of the UBI-Fights dataset [32], and 150 fight-labeled videos of the dataset presented in [33]. Although these two datasets are labeled as fight, we re-annotate them with respect to the defined classes.

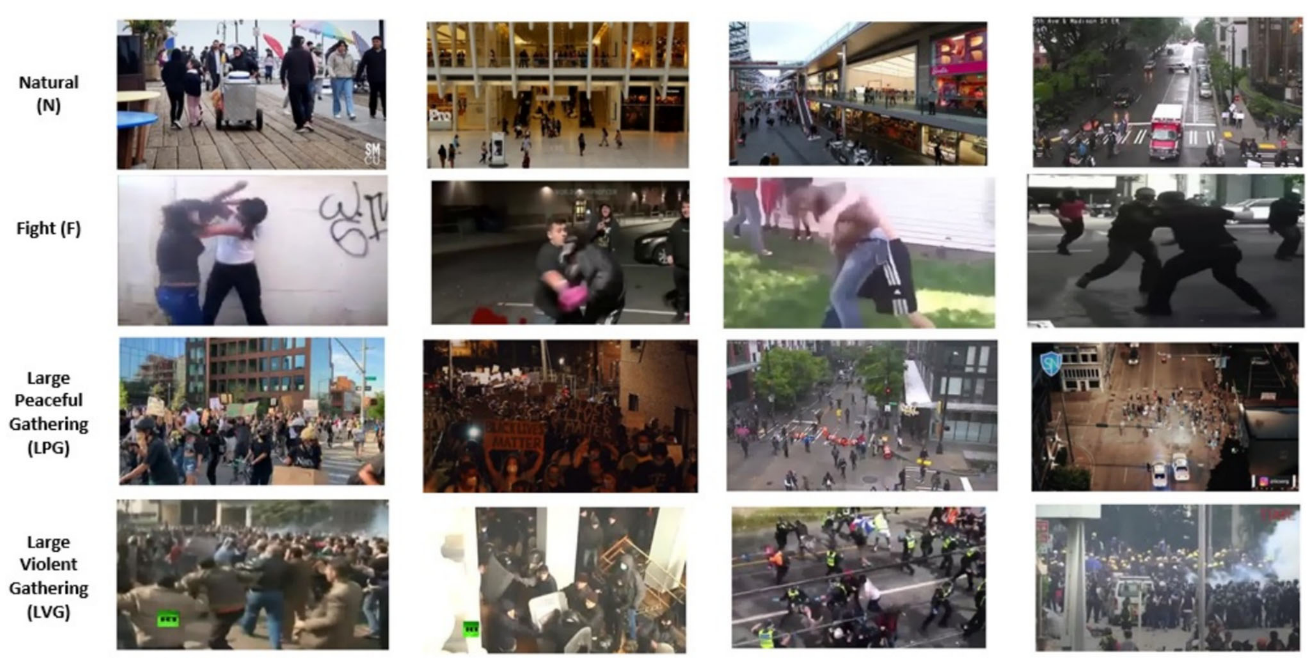

Fig. 4 Classes of crowd behavior and their sample frames from our MSV dataset

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Video preprocessing and annotation: To have consistency throughout the recorded videos, we normalize the frame rate of all the videos to an adopted frame rate of 10. Since one video may have parts of various classes at certain time intervals, it would not be appropriate to tag one label for the entire video. Instead, we identify and tag certain parts in every video that are of a specific class. All these annotations are maintained in a tabular annotation table such as Table 1. Each row of the table is for one class instance and includes video ID, start time of the segment, end time of the segment, and the corresponding class label. Time segments where no significant behaviors (such as F, LPG, or LVG) take place are noted and marked as N.

Sample extraction for model training: To preprocess the dataset for training, we consider the fixed-length samples of 2 s from all the labeled segments. With a frame rate $R = 10$ and sample duration $N = 2$ seconds, each training sample contains $R \times N = 20$ frames. For an instance of class C_i in video V_i, between times $h_i : m_i : s_i$ and $h_f : m_f : s_f$, we take all applicable frames and use a sliding window of 20 frames to create samples throughout the time range. To eliminate redundancy—because successive samples would have 19 out of 20 frames in common—we introduce a stride parameter $S = 10$. We therefore skip 10 frames between samples to weigh storage efficiency against diversity of data. Furthermore, to prevent long video segments from dominating the dataset, we impose an upper limit $E_{max} = 200$ on the number of samples for each instance. If we are able to obtain more than 200 samples from an instance, we uniformly sample 200 samples within the segment. Each sample obtained is labeled with the class C_i that it is a sample of. We have collected a total of 1,413 videos with one or more of these behavioral categories.

Ethical considerations: The dataset used in the present work comprises publicly available videos and surveillance videos downloaded from YouTube under the default YouTube license. No sensitive information was collected, and utmost care has been taken in keeping individual privacy intact. For checking potential bias, we assessed the model's performance on various subsets of data. The comparison of false negative or false positive rates among various groups or sets of circumstances showed that there was no noteworthy disparity. The model proposed is specially crafted for research purposes and for potential application in public safety deployment in ethically managed and controlled settings. It is not intended for independent use in open or uncontrolled environments. Any practical application will have to be supplemented by explicit guidelines regarding human supervision, responsibility, and properly defined constraints to avoid abuse or unethical use.

4.1.2 Benchmark datasets

In addition, we utilized the publicly available benchmark datasets such as surveillance camera fight (SCF) [33], hockey fight (HF) [34], real life violence (RLV) [35], and violent flows (VF) [5]. The SCF and HF datasets provide annotations for no-fight and fight categories, containing 300 and 1,000 videos, respectively. In contrast, the VF and RLV datasets are labeled with non-violent and violent tags, consisting of 246 videos in VF and 2000 videos in RLV. Table 2 shows the summary of the datasets used in the study.

Table 1 Sample annotation table showing five instances of behavior classes across three different videos

Video ID	Start time	End time	Class
1	00:01:30	00:02:30	LPG
1	00:04:03	00:05:31	LVG
2	00:00:25	00:00:46	N
3	00:02:27	00:02:49	N
3	00:00:01	00:00:05	F

Table 2 Datasets used in this study

Dataset	Classes	Cardinality and size
Hockey fight (HF) dataset [4]	2	1000 videos
Violent flows (VF) [5]	2	246 videos / 14 min
Surveillance camera fight (SCF) [33]	2	300 videos / 11 min
Real life violence (RLV) [35]	2	2000 videos
MSV dataset (our dataset)	4	1413 videos / 30 h

4.2 Experiment and analysis

4.2.1 Details of implementation

For training and validation, each dataset was split in a ratio of 8:2, ensuring a disjoint split. The input to our model is a video tensor. During training, we employed a frame-sampling strategy to create video tensors across various datasets. Each video was sampled into video tensors, with the length of the tensor depending on the dataset. In particular, each video tensor consists of a set of frames, such as 20 frames for MSV, SCF, and RLV, 16 frames for HF, and 11 frames for VF. To augment our training data, we applied random scaling, followed by a 224×224 random cropping process. During the inference phase, all frames undergo resizing to 256×256 and subsequently center-cropped to a final size of 224×224.

For validation, we took a different approach. Instead of using a sample selection, we processed the entire video from the validation set. This means that each video from the validation set, which constitutes 20% of the total videos in the dataset, was fed into the model one by one. This approach ensures that the model's performance is evaluated on complete video sequences, providing a more accurate assessment of its real-world applicability. To accommodate the varying lengths of videos during validation, the batch size is adjusted to one. This means that the model processes one video at a time, regardless of its length. This method is crucial for handling videos of different durations without compromising the model's performance or encountering memory issues. Through these strategies, we seek to deliver a thorough and realistic evaluation of the proposed model's performance, ensuring its robustness and reliability in real-world applications. Moreover, for training and validation, the last layer's output is set to 4 in the case of MSV and 2 for the benchmark datasets.

We utilized the residual CNN architecture for global feature backbone, APN backbone, and local feature backbone. The final layer in the classification module is a dense neural network, and the outputs are predicted using the activation function, sigmoid. The backbones are trained using stochastic gradient descent (SGD), accompanied by the following hyperparameters: learning rate annealing: cosine, momentum: 0.9. In contrast, the Adam optimizer was employed for the classification module. The training commenced with a learning rate of 0.01, for 150 epochs, utilizing full inputs. The platform for the experiments was the PyTorch framework of Python in NVIDIA Ray Tracing Texel eXtreme (RTX) 3090 Ti Graphics Processing Unit (GPU).

4.2.2 Results and discussions

The performance of the model was assessed across all the aforementioned datasets by computing the Top-1 Accuracy metric. Initially, we conducted experiments on the MSV dataset by exploring a range of configurations. We varied a manifold of *apn*s (n) and temporal depth (d), and the results of these experiments and the confusion matrices are portrayed in Table 3 and Fig. 5, respectively.

The analysis demonstrates that utilizing the residual CNN architecture with $n = 2$ and $d = 4$ achieves the highest level of accuracy and mean average precision (mAP) compared to other configurations. To validate the choice of using two dynamic attention maps $n = 2$, we conducted a one-way ANOVA analysis on classification accuracy and mAP across different values of $n \in 1, 2, 3$. For accuracy, the ANOVA test revealed a statistically significant effect of the number of attention maps, with $F(2, 3) = 12.662$, $p = 0.0345$, indicating that the

Table 3 Accuracy (%) and mAP (%) for varying *apn*s (*n*) and temporal depth (*d*) in MSV Dataset

n	*d*	Accuracy (%)	mAP (%)
1	4	77.2	79.5
1	8	75	77.3
2	4	**85**	**86.3**
2	8	82.28	83.6
3	4	82.54	82.9
3	8	81	81.4

The highest values are shown in bold

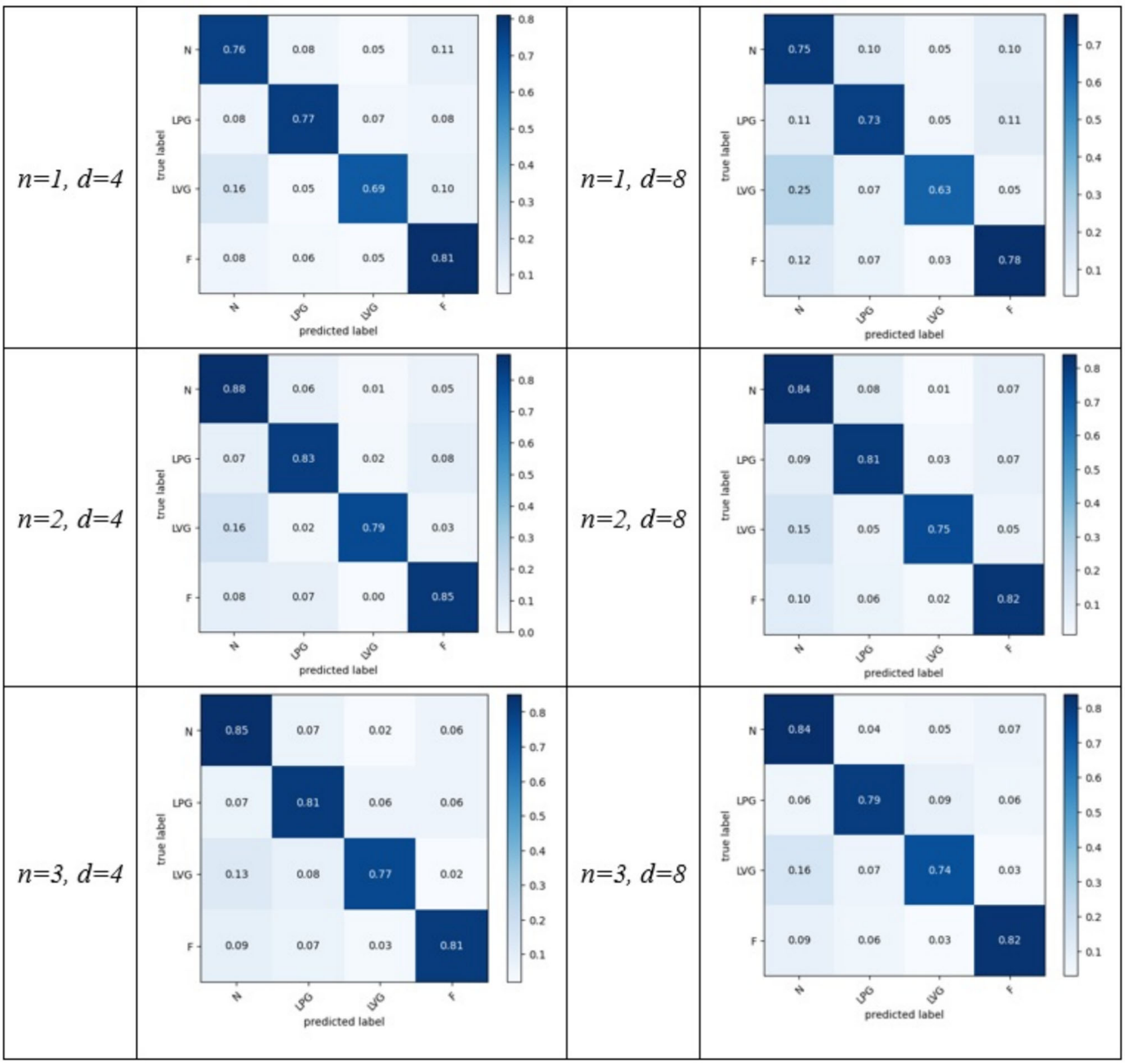


Fig. 5 Confusion matrices portraying the per-class accuracy in the MSV dataset for various *n* and *d*

differences in performance across the groups are unlikely due to chance. For mAP, the result was marginally significant, with $F(2,3) = 9.013$, $p = 0.0539$, suggesting a meaningful trend favoring $n = 2$, although just above the conventional 0.05 threshold. These results empirically support the claim that using two attention maps yields superior and more consistent performance.

Furthermore, this outcome suggests that simulating multiple attention capabilities, similar to human visual attention, ensures more accurate recognition than focusing on a single attention point ($n = 1$). Interestingly, the results for $n = 3$ are slightly lower than for $n = 2$. These findings align with the principles of human visual attention. Humans typically focus on a Limited number of areas within their visual field at any given time. Research indicates that humans can effectively manage and switch attention between a few focal points, typically around 2-3 [36, 37]. This limited attentional capacity allows for detailed processing of multiple regions while maintaining overall efficiency. For instance, focusing on $n = 2$ attention points mirrors the human ability to simultaneously process a couple of key areas, leading to optimal performance. Increasing the number of attention points to $n = 3$ or more might introduce a cognitive load that slightly reduces efficiency, as humans also experience diminishing returns when spreading attention too thin across multiple areas [38].

We also calculated the time of inference and throughput of the model in the MSV dataset, and the results are displayed in Fig. 6. The inference time is under 1 s for both GPU and central processing unit (CPU), whereas the throughput for a batch size of 4 is 79.1 frames per second (fps) for GPU and 3.8 fps for CPU. The CPU for these computations was Advanced Micro Devices (AMD) Ryzen Threadripper 3960X 24-core processor, while the GPU was NVIDIA GeForce RTX 3090 Ti. The model demonstrates promising inference performance, showing strong potential for real-time deployment in practical surveillance settings. With GPU-based throughput reaching 79.1 frames per second and inference time under one second, the model is well-suited for moderate to large-scale surveillance systems requiring responsive analysis. These results indicate that, with dedicated GPU acceleration, the framework can efficiently handle real-time video streams across various scenarios. While performance varies on CPU-only configurations, this opens exciting opportunities for future optimization. Ongoing efforts are focused on further enhancing efficiency through lightweight model architectures, paving the way for broader deployment across city-scale and resource-constrained environments. These developments position the model as a strong candidate for scalable, high-impact applications in intelligent video surveillance.

Furthermore, the results from our experiments on the MSV dataset paved the way for further investigations on the benchmark datasets, particularly for the configuration $n = 2$ and $d = 4$. The summarized outcomes of these evaluations are displayed in Fig. 7. The videos in the MSV dataset, along with the other datasets used in this study, include a wide range of crowded scenes captured under various environmental conditions, including occlusion, low lighting, and different weather scenarios. The performance results indicate that the proposed model is capable of effectively perceiving and analyzing spatiotemporal patterns related to violent and non-violent behaviors across diverse visual conditions, highlighting its practical robustness. However, certain extreme conditions—such as very poor illumination, significant motion blur, or densely populated scenes with persistent occlusions—remain challenging for any vision-based system and represent opportunities for further model enhancement. Likewise, situations involving rapid camera motion, overlapping actions, or non-standard viewpoints offer potential for extending the model's generalization capabilities. As part of future work, we aim to

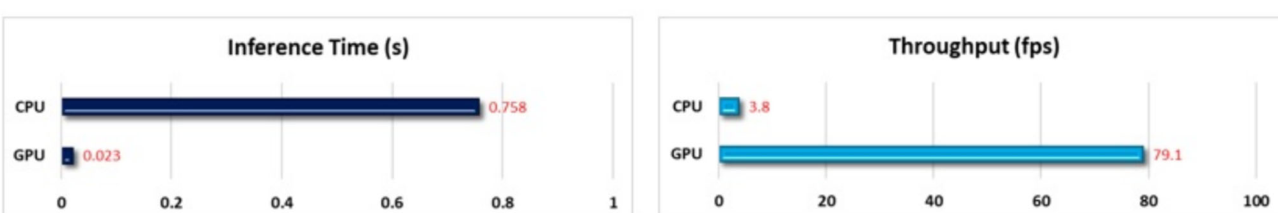


Fig. 6 Inference time in seconds and throughput in fps of the model in GPU and CPU using the MSV dataset

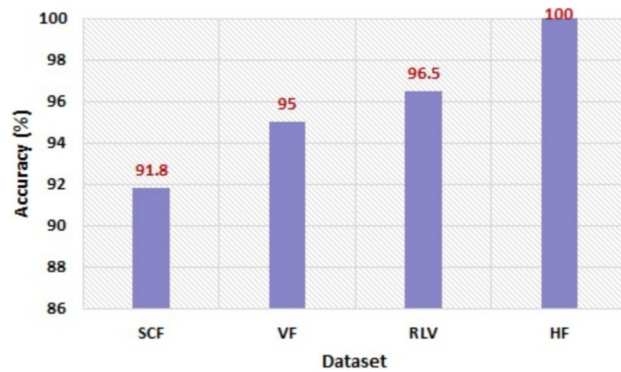


Fig. 7 Top-1 accuracy (%) in the benchmark datasets for $n = 2$ and $d = 4$

improve resilience under these conditions by incorporating motion stabilization techniques, targeted data augmentation for edge cases, and multimodal inputs such as depth, thermal imagery, or audio cues.

Moreover, since many videos were sourced from public platforms such as YouTube, the datasets naturally include multiple camera angles and perspectives, enhancing the model's robustness to viewpoint variations. Regarding class distribution, all benchmark datasets used in the experiments are class-balanced. However, the MSV dataset exhibits a moderate imbalance across its four categories: N (523 videos), LPG (405 videos), LVG (113 videos), and V(372 videos). While this imbalance reflects real-world distributions to some extent, it may introduce bias toward more frequently occurring classes. In future work, we aim to further refine dataset balance through targeted data collection and augmentation strategies and evaluate model fairness across frequent and less-represented categories.

4.2.3 Comparison with existing approaches

Top-1 accuracy was computed to compare the model with existing approaches in various datasets. The results are given in Table 4. Various configurations of *apns* (n) with different depths (d) were tested across our MSV dataset and benchmark datasets. This architecture allowed us to effectively analyze how multiple attention points influence the recognition accuracy compared to a single attention point.

Our approach with $n = 2, d = 4$ consistently provided better performance than existing methods in the Literature, demonstrating that simulating multiple attention capabilities similar to human visual attention improves recognition accuracy. This aligns with research indicating that humans can effectively manage attention between 2-3 focal points for optimal performance [36, 37]. Increasing the number of attention points to $n = 3$ slightly reduces efficiency, likely due to the cognitive load of spreading attention too thinly [38]. The results reveal the potency of the model in capturing spatiotemporal patterns associated with fight and violent scenarios across different locations.

Comparison with transformer models: We specifically selected transformer-based methods such as vision transformer (ViT) [60], video vision transformer (ViViT) [63], video transformer [66], and video swin transformer [67] for comparison, as their classifications rely on modeling the relationships (attention) between spatial and temporal patches in the frames. Other methods were chosen because they are widely used for violent detection in video data. The classes of benchmark datasets considered in this work represent no-fight/fight and no-violence/violence scenarios. Moreover, there were fewer people interacting with clearer visuals in the HF dataset, whereas the VF dataset provides more unique patterns for non-violent and violent behavior, which would contribute to the model learning and generalizing well. On the other side, greater diversity and complexity result in variance of camera angles, lighting, and even crowd dynamics at play that challenge a model's performance in the SCF and RLV Datasets. However, our approach invariably outperformed the current methods even in these richer settings.

While the proposed model demonstrates consistently strong performance across multiple benchmark and real-world datasets, there remain exciting opportunities for further enhancement, particularly in the area of

Table 4 Comparison of accuracy (%) in HF, SCF, VF, RLV, and MSV datasets

Methods	HF	SCF	VF	RLV	MSV
Bi-LSTM + VGG16 [33]	–	52	-	-	-
LSTM + Xception CNN [33]	–	55	–	–	–
LSTM + VGG16 [33]	–	61.67	–	–	–
Bi–LSTM + Xception CNN [33]	–	63	–	–	–
Attention + Bi–LSTM+Xception CNN [33]	–	68	–	–	–
Aktı et al. [33]	–	72	–	–	–
Ullah et al. [39]	–	75.9	–	–	–
Hassner et al. [5]	82.9	–	81.3	–	–
Local Binary Tracklets(LBT) [40]	–	–	81.90	–	–
Mousavi et al. [41]	–	–	82.30	–	–
3DCNN [29, 42]	–	87.2	–	–	–
ViF+Oriented ViF [43]	87.5	–	–	–	–
CNN+LSTM [29, 35]	–	–	–	88.8	–
Temporal Fusion CNN + LSTM [44]	–	–	–	91	–
Li et al. [45]	90.5	81.3	92.41	–	–
AR-Net [46]	92	76.4	94.2	–	–
I3D–Conv Net [47]	93.4	–	–	–	–
Three streams + LSTM [48]	93.9	–	–	–	–
MoSIFT+KDE [49]	94.3	–	–	–	–
I3D [50]	94.5	84.6	88.5	–	78.5
Xu et al. [49]	–	–	89.05	–	–
Gao et al. [51]	–	–	90.17	–	–
3DCNN+SVM [52]	–	–	90.6	–	–
Varghese et al. [53]	–	–	92.9	–	–
Dictionary learning [54]	–	–	93.19	–	–
Vijeikis et al. [55]	94.7	70.8	–	–	–
ResNet50 [56]	95	85	93.8	–	–
Su et al. [57]	96.8	–	–	–	–
Hanson et al. [9]	96.54	–	92.18	–	–
Convolutional LSTM [8]	97.1	–	–	–	–
Obregón et al. [58]	97.4	–	–	–	–
Lin et al. [59]	97.5	78.2	94	–	–
ViT [60]	97.9	89.2	94.1	92.5	79
Ciampi et al. [61]	98	83.2	–	–	–
VD–Net [62]	98	90.6	94	–	–
ViViT [63]	98.5	90.8	94.5	93	80.5
Qaraqe et al. [64]	98.5	81.51	98.5	–	**88.89**
AdaFocusV2 [28]	–	–	–	–	82
R(2+1)D [65]	–	–	–	–	83.23
ResNet3D [65]	–	–	–	–	83.74
Video Transformer [66]	98.8	91.5	94.5	93	83.8
Swin Transformer [67]	98.8	91.7	94.7	94.5	83.9
Hachiuma et al. [31]	–	–	94.7	–	–
Abdali et al. [68]	–	–	–	96.25	–
Proposed (n=1, d=4)	92.5	84.6	88.1	89.4	77.2
Proposed (n=1, d=8)	90.2	83.4	85.4	85.6	75
Proposed (n=2, d=4)	**100**	**91.8**	**95**	**96.5**	85
Proposed (n=2, d=8)	99.5	90.2	94.8	95.6	82.28

Table 4 (continued)

Methods	HF	SCF	VF	RLV	MSV
Proposed (n=3, d=4)	**100**	91	94.5	94.7	82.54
Proposed (n=3, d=8)	98.8	89.95	93.6	93	81

The highest values are shown in bold

optimization. The model relies on gradient-based optimizers such as stochastic gradient descent (SGD) and its variants (e.g., Adam). These have been extensively validated in deep learning and offer fast convergence and scalability, especially in complex structures such as spatiotemporal attention models. However, as is with the majority of gradient-based approaches, they optimize locally and do not inherently ensure convergence to a global optimum, particularly under very non-convex loss landscapes. In future studies, this aspect opens room for the use of metaheuristic optimization techniques, which are famous for their global search capability and local minimum avoidance [69].

4.2.4 Qualitative analysis

The visualization results are illustrated in Fig. 8, with green boxes indicating the areas of the scene where the model selects attention maps. The figure demonstrates the model's ability to identify crucial regions within the video that correctly classify violent behavior. These scenarios were sourced from the MSV dataset, where fine-grained recognition is essential to distinguish various violent crowd behaviors. The proposed model accurately identifies behaviors occurring at different locations throughout the video, including those at the video's end, showcasing its effectiveness in multi-attention fine-grained video recognition.

Visualizations are also presented as heatmaps created using gradient-weighted class activation mapping (Grad-CAM) [70]. These visual explanations taken from the MSV dataset are illustrated in Fig. 9, which showcases the location of attention for $n = 2$ and $n = 3$. The results are obtained by superimposing a visualization layer after the fully connected layer in the classification module. The heatmap localization highlights the key areas in the frames where the attention maps are concentrated for class prediction.

4.2.5 Ablation studies

Effect of coarse global feature backbone: The proposed model utilized the fusion of coarse global feature vector ($\mathbf{f_g}$) with local features extracted from attention backbone ($\mathbf{f_l}$) to derive the final classification result. To further understand the impact of these global maps, experiments were done on the MSV dataset with and without the aggregation of $\mathbf{f_g}$ with $\mathbf{f_l}$. The results are demonstrated in Fig. 10. The introduction of $\mathbf{f_g}$ to the model serves the purpose of enhancing the utilization of spatial information in the classification process. As evident from the figure, the inclusion of supplementary spatial details notably contributes to the robust stratification of violent patterns.

Experiments with multiple feature backbones: Our model primarily relies on residual CNN for global, attention, and local backbones. In addition to this, on the MSV dataset, we employed the transformer networks and the LSTM+Attention network as in [19, 23, 24] to extract global and attention proposal features. Figure 11 provides the results of these experiments. Among the architectures, R(2+1)D consistently achieves the highest accuracy, particularly at n=2, d=4, where it reaches 85%. Increasing n from 1 to 2 generally improves performance across all models, suggesting that incorporating more temporal context enhances classification accuracy.

5 Conclusion

Violence detection in videos has gained significant promise in practical applications, driven by the increase in video content in recent years. However, many existing approaches treat violence detection as a straightforward video classification task, often falling short in identifying violent actions that tend to occur within a limited spatiotemporal region within the video. This challenge is addressed by developing a novel temporal–spatial framework with 3D dynamic attention maps, designed for fine-grained violence recognition in videos. This model draws inspiration from the visual attention capabilities of humans, allowing it to focus on multiple events within a scene. The model incorporates spatiotemporal dynamic *apn*s, which play a pivotal role in comprehending temporal relationships between video frames and dynamically allocating attention to informative spatial and temporal segments. The 3D volume of these *apn*s not only indicates where and when to direct attention but also determines the duration over which inferences should be made, leading to an overall enhancement in performance. Experiments conducted on diverse datasets demonstrate its efficiency in interpreting complex actions and scenes, enabling precise recognition. This innovative approach holds great promise for diverse applications in computer vision, paving the way for more accurate and insightful video analysis.

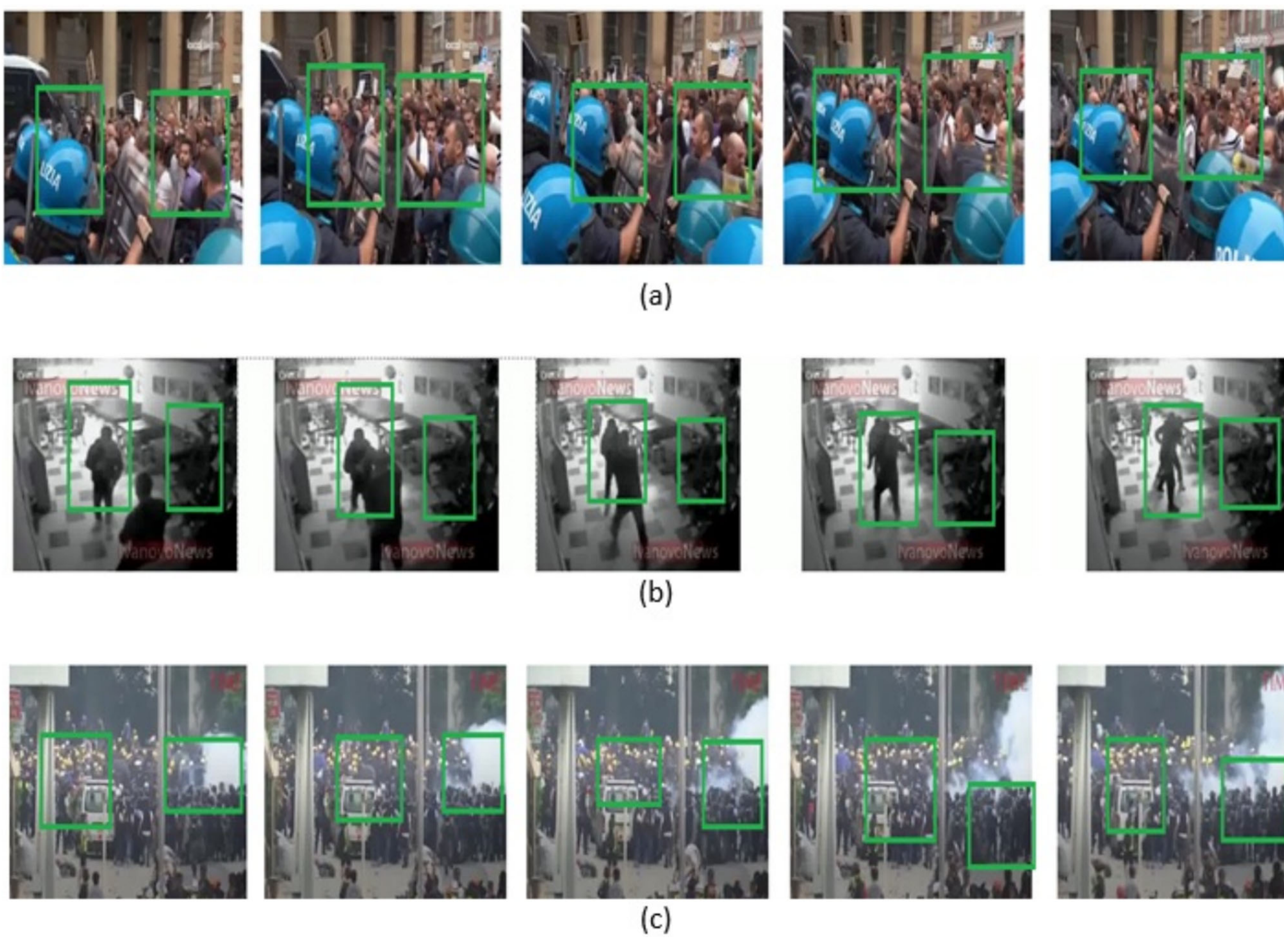


Fig. 8 Visualization results of the proposed model for $n = 2$. **a** Violence emerges in a large crowd only toward the end of the scene. Our model correctly identifies the scene where the other models fail. **b** Similar to **a**, the fight scene is at the end of the video, leading to the classification as a *Natural* scene instead of *Fighting* by other models except the proposed model. **c** A large, violent gathering where violent locations can be correctly identified by the proposed model

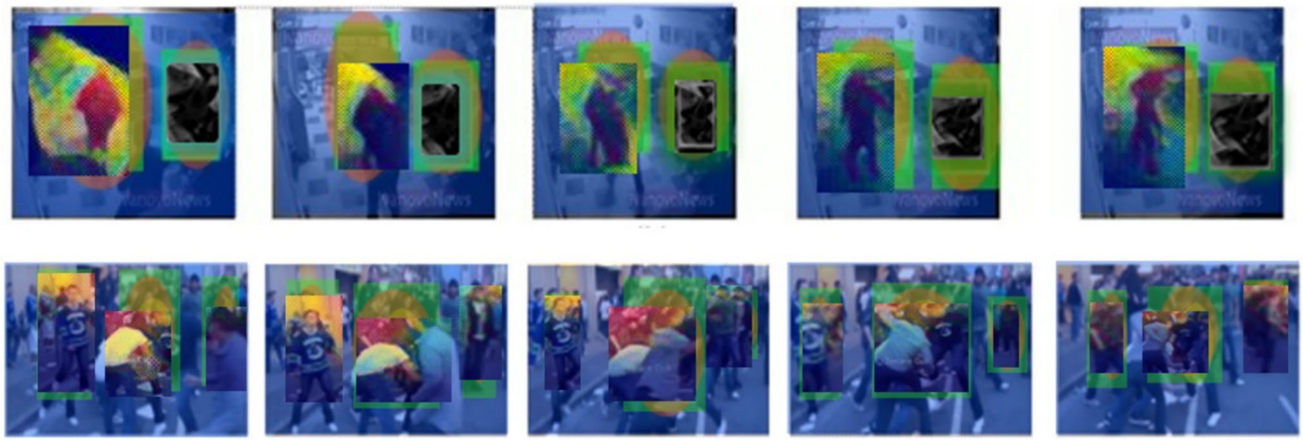
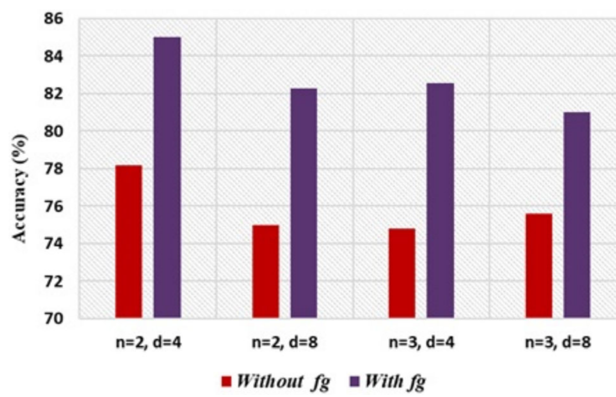
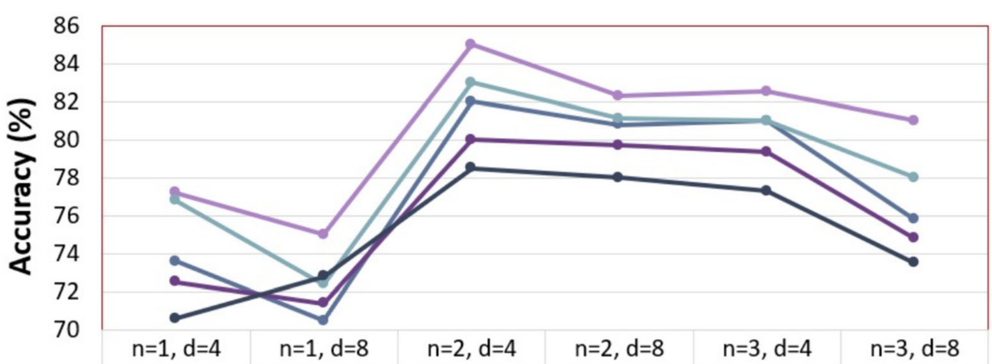


Fig. 9 Visual explanation of attention maps using Grad-CAM. First row: $n = 2$, Second row: $n = 3$. The results are taken from the MSV dataset

Fig. 10 Impact of $\mathbf{f_g}$ in the proposed model on the MSV dataset





	n=1, d=4	n=1, d=8	n=2, d=4	n=2, d=8	n=3, d=4	n=3, d=8
R(2+1)D	77.2	75	85	82.28	82.54	81
ResNet3D	73.6	70.5	82	80.8	81	75.8
Swin Transformer	76.8	72.4	83	81.12	81	78
ViT	72.5	71.4	80	79.7	79.35	74.8
LSTM + Attention	70.6	72.8	78.5	78	77.3	73.5

Fig. 11 Comparison of accuracy (%) for multiple *apn*s (n) and temporal depth (d) using different feature backbones in MSV dataset

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Data availability The datasets generated during the current study are available from the corresponding author upon reasonable request.

Code availability The code of the current study is available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare they have no conflict of interest to report regarding the present study.

Ethics approval No ethics approval was required for the study.

Consent for publication All authors have approved the manuscript and agree with its publication.

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